# INORGANIC CUME: SYMMETRY

April 17, 2009

Because of the diversity of structures in inorganic chemistry, point group notation ($C_{2v}$ etc.) is a valuable way to describe molecular structures.

For each of the following give a <u>sketch</u> of the molecule and give the <u>point group notation</u> for the molecule:

$SO_3$

$SO_3^{-2}$

$IF_3$

$XeF_4$

$CO_2$

Ferrocene

$B_6H_6^{-2}$

White phosphorus

$[Pt(CN)_4]^{-2}$

$IF_7$

Ammonia

fac-$[Co(NH_3)_3Cl_3]$ (ignore the H's)

each of the three isomers of $PCl_2F_3$

each of the two isomers of $[CrCl_4(CN)_2]^{-3}$